Simulation Arena

Deal Approval & Margin Balancing

Approve 18% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $4.30M of expected profit.

Seed 42 | Run run_ec6e7301dc7c | 170,000 scenarios tested | Monte Carlo simulation with an empirica… | 21 Sep 2026, 23:38

Same seed, same numbers - this run can be reproduced exactly.

How this works

For each discount level we ran thousands of 'what-if' worlds. In each one the buyer's odds of saying yes wobble a little, and your delivery cost wobbles a little. We then ask the only question that matters: if you offered this discount again and again, how much profit would you bank on average? That is the expected profit - the chance of winning multiplied by the profit you keep. Because those worlds are a sample, we also work out how much the answer itself would wobble if we ran it again. Any discount whose expected profit is within twice that wobble of the best - or within 1% of the best profit, whichever is wider - is treated as giving the same answer. That is the range where you have genuine room to negotiate.

The numbers that matter

BEST DISCOUNT TO OFFER
18%
This is where the extra wins are still worth more than the margin given away.
vs rep asked for 25%

EXPECTED PROFIT AT THAT DISCOUNT
$18.87M
Average profit if you made this call many times over.

DISCOUNT RANGE YOU CAN APPROVE WITHOUT LOSING MONEY
18%
Only one discount stands out - the model cannot tell any neighbouring discount apart from it. The model's own margin of error here is about $429k, so treat this as the best single number - the next discount along already costs real money.
vs rep asked for 25% - outside the range

WHERE IT STARTS TO GENUINELY COST YOU
20%
At 20% the expected profit falls $396k below the best. That gap is wider than the model's own margin of error, so the loss is real.
vs costs about $396k

EXPECTED PROFIT IF YOU CAVE IN
$14.57M
What 25% actually earns you.
vs -$4.30M vs the optimum

PROFIT GIVEN AWAY
$4.30M
Buying +15 points of win probability costs you this much.

ODDS OF WINNING AT THE BEST DISCOUNT
77.5%
The chance the deal lands.

WORST CASE (1-IN-20)
$0
In an unlucky run of 20 deals, this is roughly the profit you would see.

How we got there

1. The two forces fighting each other

Every point of discount does two things: it makes the buyer a bit more likely to say yes, and it takes $1.12M straight off your profit. Early on the first effect wins. Later, the second one does.

Evidence: Cost to deliver $67.91M on $111.58M of list value

2. What the buyer's behaviour looks like

They start around 35.0% likely to buy at full price and never become certain - the curve flattens near 95.0%. Around 15% discount is where they are most persuadable.

Evidence: Curve source: 450 of your past deals

3. The sweet spot

18% discount gives the highest expected profit at $18.87M, with a 77.5% chance of winning.

Evidence: peak of 17 levels tested

4. The range where the answer is the same

18% is the only discount that stands out; the model cannot tell any neighbouring discount apart from it. The next level along already costs $396k.

Evidence: margin of error $429k; safe range 18%

5. What the requested discount really costs

Moving from 18% to 25% buys +15 points of win probability but costs $4.30M of expected profit - more than the model's margin of error, so the loss is real.

Evidence: $14.57M vs $18.87M

Charts

Where profit peaks

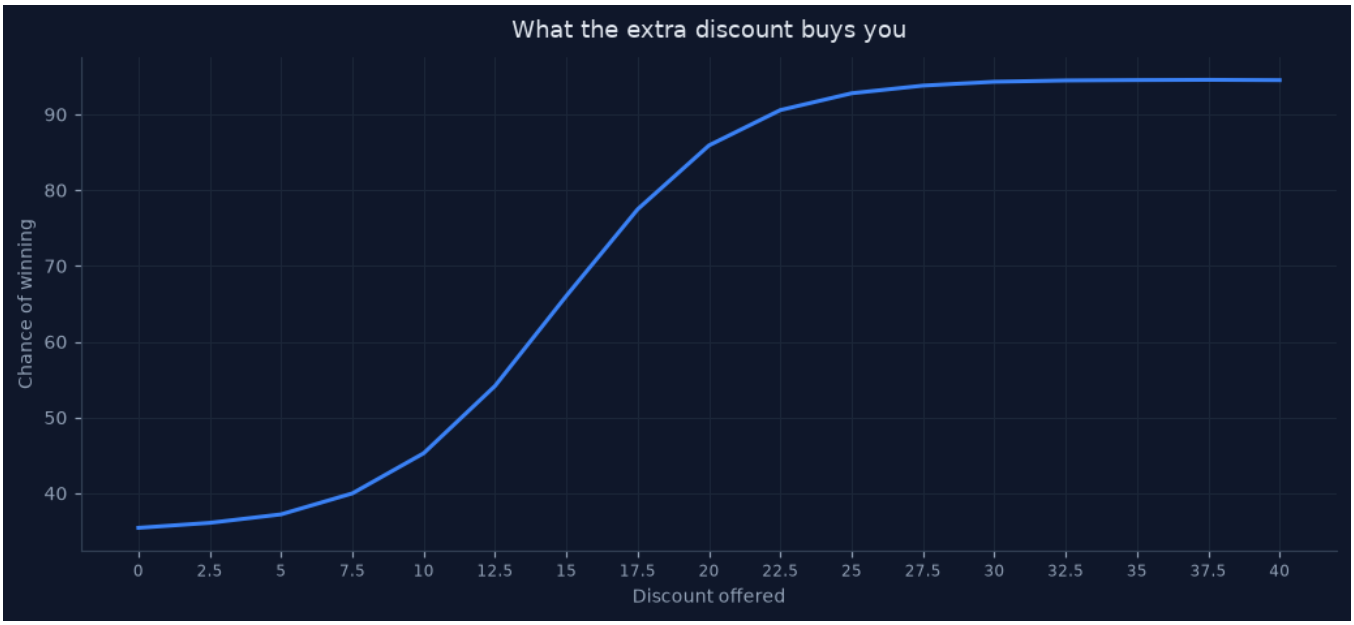
Everything from 18% gives the same answer within the model's margin of error; the peak is 18%



What this shows: Grey is what you keep per won deal - it only falls. Green is what you actually expect to earn - it rises, peaks, then falls. The amber dashed line is the best profit minus the model's own margin of error: every discount whose green line stays above it is the same answer, which is the 18% range. Past that you are genuinely paying for the win.

What the extra discount buys you

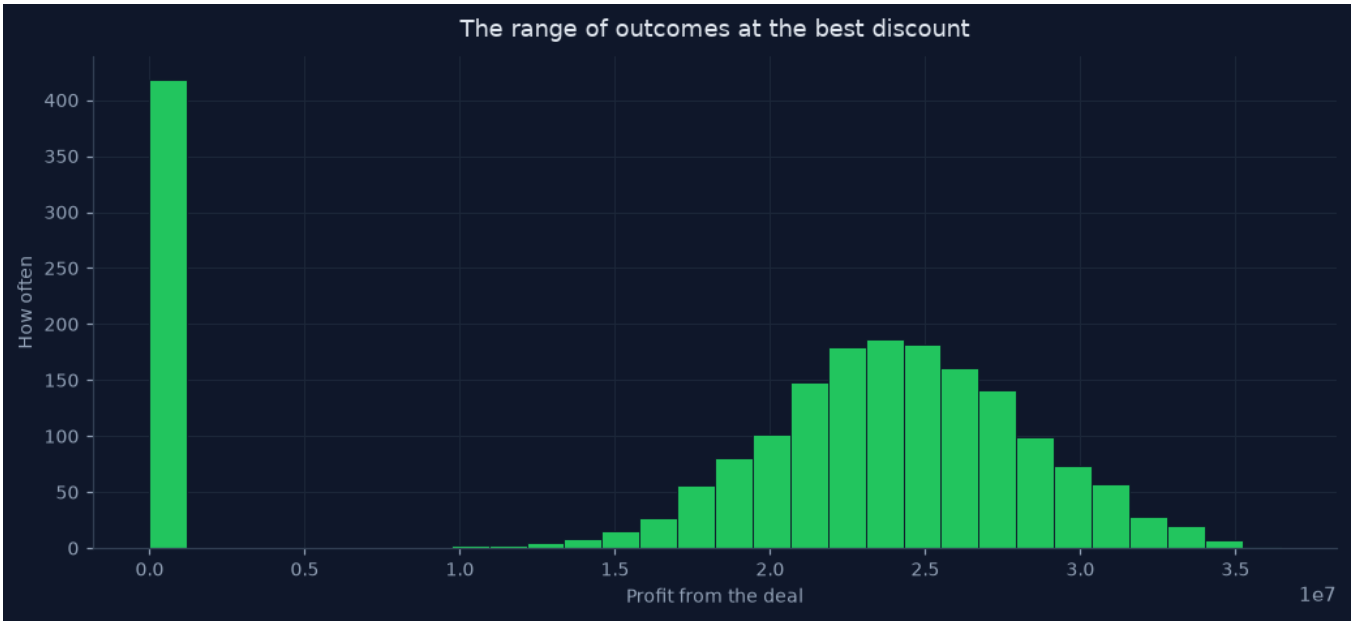
Curve learned from 450 of your past deals



What this shows: Past the knee of this curve, extra discount buys almost no extra certainty - but keeps costing margin. That is the trap.

The range of outcomes at the best discount

10,000 simulated worlds at 18% discount



What this shows: The cluster at zero is the deals you lose. The spread on the right is the profit you make when you win.

Tables

Every discount level tested

Expected profit = chance of winning x profit kept. Highlighted row is the winner.

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
0.0%	$111,585,000	$43,671,451	35.4%	$15,646,106	$0	0% chance
2.5%	$108,795,375	$40,927,206	36.1%	$14,631,398	$0	0% chance
5.0%	$106,005,750	$38,096,443	37.2%	$14,300,515	$0	0% chance
7.5%	$103,216,125	$35,284,778	40.0%	$14,221,904	$0	0% chance
10.0%	$100,426,500	$32,503,199	45.3%	$14,617,138	$0	0% chance
12.5%	$97,636,875	$29,851,295	54.2%	$16,094,262	$0	0% chance

Discount	Price after discount	Profit if won	Win chance	Expected profit	Worst case	Breaches floor?
15.0%	$94,847,250	$26,963,322	66.1%	$17,832,984	$0	0% chance
17.5%	$92,057,625	$24,252,399	77.5%	$18,865,622	$0	0% chance
20.0%	$89,268,000	$21,410,297	85.9%	$18,469,167	$0	0% chance
22.5%	$86,478,375	$18,588,255	90.6%	$16,905,092	$0	2% chance
25.0%	$83,688,750	$15,775,567	92.8%	$14,568,704	$0	8% chance
27.5%	$80,899,125	$12,959,049	93.8%	$12,162,348	$0	21% chance
30.0%	$78,109,500	$10,107,675	94.3%	$9,528,412	$0	43% chance
32.5%	$75,319,875	$7,392,684	94.5%	$6,947,155	$0	66% chance
35.0%	$72,530,250	$4,558,265	94.5%	$4,330,672	-$1,971,519	85% chance
37.5%	$69,740,625	$1,898,703	94.6%	$1,801,412	-$4,584,397	95% chance
40.0%	$66,951,000	-$957,538	94.5%	-$910,607	-$7,512,832	99% chance

Row 8 maximises expected profit.

The decision in one line

What changes if you approve the discount being asked for.

Column 1	Best (18%)	Asked (25%)	Difference
Discount	17.5%	25.0%	+7.5 pts
Win chance	77.5%	92.8%	+15.3 pts
Profit if won	$24,252,399	$15,775,567	-$8,476,833
Expected profit	$18,865,622	$14,568,704	-$4,296,918
Range where the answer is the same	18%	outside it	margin of error $429k

The range where the answer is the same

Every discount we tested, how far it falls short of the best, and whether that shortfall is real money or just the model's own margin of error.

Discount	Expected profit	Short of the best	Inside the range?	What it means
0.0%	$15,646,106	$3,219,516	Outside	costs you $3.22M
2.5%	$14,631,398	$4,234,224	Outside	costs you $4.23M
5.0%	$14,300,515	$4,565,107	Outside	costs you $4.57M
7.5%	$14,221,904	$4,643,719	Outside	costs you $4.64M
10.0%	$14,617,138	$4,248,484	Outside	costs you $4.25M
12.5%	$16,094,262	$2,771,360	Outside	costs you $2.77M
15.0%	$17,832,984	$1,032,638	Outside	costs you $1.03M
17.5%	$18,865,622	$0	Inside	safe to approve
20.0%	$18,469,167	$396,455	Outside	costs you $396k
22.5%	$16,905,092	$1,960,530	Outside	costs you $1.96M
25.0%	$14,568,704	$4,296,918	Outside	costs you $4.30M
27.5%	$12,162,348	$6,703,274	Outside	costs you $6.70M
30.0%	$9,528,412	$9,337,210	Outside	costs you $9.34M
32.5%	$6,947,155	$11,918,467	Outside	costs you $11.92M
35.0%	$4,330,672	$14,534,951	Outside	costs you $14.53M
37.5%	$1,801,412	$17,064,210	Outside	costs you $17.06M
40.0%	-$910,607	$19,776,229	Outside	costs you $19.78M

Highlighted rows are the range where the answer is the same (18%) - within about $429k of the best.

Recommended discount per deal (top 25)

The same logic applied to each deal individually.

Deal	Customer	Value	Margin	Best discount	Expected profit
D-0413	Tyrell Holdings	$1,922,000	45.1%	20%	$414,250
D-0251	Globex Energy	$1,663,000	42.0%	18%	$316,043
D-0084	Basalt Motors	$1,395,000	39.2%	18%	$234,813
D-0312	Verity Group	$1,383,000	42.7%	18%	$270,341
D-0214	Vehement Energy	$1,309,000	30.4%	0%	$141,078
D-0091	Wayne Retail	$1,104,000	38.4%	18%	$178,980
D-0103	Hooli Systems	$1,084,000	31.5%	0%	$121,056
D-0307	Kestrel Bank	$1,025,000	35.9%	18%	$146,295
D-0048	Duff Group	$995,000	30.3%	0%	$106,884
D-0053	Ironwood Systems	$985,000	38.6%	18%	$161,216
D-0419	Vehement Media	$984,000	34.4%	18%	$128,994
D-0203	Vandelay Media	$922,000	35.8%	18%	$130,879
D-0355	Initech Logistics	$858,000	34.7%	18%	$114,473
D-0269	Gringotts Group	$856,000	28.3%	0%	$85,883
D-0302	Massive Motors	$827,000	33.4%	18%	$101,998
D-0068	Tessera Group	$820,000	38.5%	18%	$133,574
D-0428	Prestige Partners	$811,000	45.9%	20%	$180,367
D-0195	Vandelay Chemicals	$798,000	37.3%	18%	$122,562
D-0028	Initech Logistics	$756,000	48.2%	20%	$183,065
D-0364	Verity Systems	$743,000	39.6%	18%	$127,371
D-0430	Prestige Holdings	$738,000	33.8%	18%	$93,311
D-0323	Gringotts Energy	$721,000	32.2%	0%	$82,307
D-0004	Hooli Freight	$697,000	35.4%	18%	$96,777
D-0444	Cyberdyne Freight	$693,000	39.1%	18%	$116,112
D-0317	Cobalt Labs	$677,000	39.7%	18%	$116,582

What we recommend

Approve 18% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $4.30M of expected profit.

Risk Low 25 out of 100 | Confidence: High

Why

• Only 18% stands out from the rest - the model cannot tell any neighbouring discount apart from it.

• Expected profit peaks at 18% ($18.87M).

• The requested 25% delivers $14.57M - $4.30M less than the peak, which is more than the model's margin of error.

• Win probability only improves +15 points for that giveaway.

What to do

1. Approve 18% as the standard concession - it is the only discount the model can tell apart from the rest.

2. Above 18% (the top of the safe range), require VP sign-off: the 25% being asked for costs about $573k per point of discount.

3. If this quarter must not misfire, 0% is the safer call: it maximises the bad case rather than the average.

The trade-off: You are trading $4.30M of expected profit for +15 points of win probability.

What we assumed and what to watch

Assumptions

• List value $111,585,000; cost to deliver $67,906,095 at the 39.1% standard margin your file implies.

• Consumption effects: none - your file has no tier accelerator, burst or rollover columns, so this is a straight price-versus-margin question.

• Win probability curve: from 35.0% at full price toward a ceiling of 95.0%, halfway at 14.8% discount (source: 450 of your past deals).

• Win probability uncertainty: plus or minus 5 points.

• Cost volatility: plus or minus 6% of cost.

• Margin floor: 12.0%.

• Two discounts are treated as giving the same answer when the gap between their expected profits is smaller than twice the model's own margin of error at both discounts added together (about $429k at the best one), or smaller than 1% of the best profit - whichever is larger.

• The range where the answer is the same is 18%, taken from the 17 discount levels we tested.

• 10,000 simulations per discount level; seed 42 (re-running gives identical numbers).

• Deal / quote ID -> your 'deal_id' column

• Customer name -> your 'customer' column

• Deal value (list price) -> your 'deal_value' column

• Cost to deliver -> your 'cost' column

• Standard margin (%) -> your 'margin_pct' column

• Discount given (%) -> your 'discount_pct' column

• Won? (1/0) -> your 'won' column

• Hit a consumption tier accelerator -> not in your file; not used

• Unmetered burst used -> not in your file; not used

• Rollover window -> not in your file; not used

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
0.00s	prepare	Discount given (%) -> your 'discount_pct' column
0.00s	prepare	Won? (1/0) -> your 'won' column
0.00s	prepare	Hit a consumption tier accelerator -> not in your file; not used
0.00s	prepare	Unmetered burst used -> not in your file; not used
0.00s	prepare	Rollover window -> not in your file; not used
0.01s	prepare	Read 450 records into the model.
0.01s	simulate	Running the simulation…
0.01s	read	Reading the deal you gave us...
0.01s	read	Found 450 deal(s) worth $111.58M at an average 39.1% margin.
0.01s	prepare	Cost to deliver is about $67.91M.
0.01s	prepare	Working out how much a discount really moves the buyer...
0.03s	prepare	Learned the buyer's behaviour from 450 past deals (they flip hardest around 14.8% discount).
0.03s	simulate	Testing 17 discount levels, 10,000 outcomes each (170,000 scenarios in total)...

Time	Stage	What happened
0.03s	simulate	Simulated 11% of the discount levels...
0.03s	simulate	Simulated 23% of the discount levels...
0.03s	simulate	Simulated 35% of the discount levels...
0.03s	simulate	Simulated 47% of the discount levels...
0.04s	simulate	Simulated 58% of the discount levels...
0.04s	simulate	Simulated 70% of the discount levels...
0.04s	simulate	Simulated 82% of the discount levels...
0.04s	simulate	Simulated 94% of the discount levels...
0.04s	simulate	Simulated 100% of the discount levels...
0.04s	summarize	All discount levels tested. Looking for the sweet spot...
0.04s	summarize	Checked how much the answer would wobble if we ran it again - the range where the answer is the same is 18%.
0.07s	summarize	Done. Best discount is 18% ($18.87M expected profit); the range where the answer is the same is 18%.